S1 Table. Agent roster

Agent roster. All agent roles in the BURROW project (Table 4 of the main text lists roles 1–25; roles 26–27 produced this manuscript), with the model used, the main files each role produced (paths in the project repository), and how the role's output was checked by an agent that did not produce it. Opus, Claude Opus 5.5; Sonnet, Claude Sonnet 5.5.

#	Role	Model	Main outputs	Independent check
–	PI agent	Opus	DESIGN.md, docs/PREREG.md amendments, RESEARCH_TREE.md, LESSONS.md, all commits	Design reviews (6–13); process retrospectives (22–23)
1	Literature reviewer (2 research sub-agents)	Opus	LITERATURE.md	Design reviewers for prior work (7, 12)
2	Data acquisition	Opus	DATA.md; download and verification scripts; MMseqs2 and Foldseek set-up	Checksums; leakage auditor (19)
3	Baselines engineer	Sonnet	P2Rank 2.4.2/2.5.1, fpocket 4.0.3/4.2.3, fpocket+PRANK and 8 optional tools; baseline runs; P2Rank retrained on our data; apo-dev; PocketMiner and CryptoBench-holo analyses	Reproduction gate against published LIGYSIS rows; test verifier (21)
4	Evaluation engineer	Opus	Evaluation harness and metrics; LIGYSIS reproduction; cluster bootstrap; gated test runner and ledger; preregistered analysis	Reproduction of 33/33 published variants; test verifier (21) with no project imports
5	Geometry and HPC engineer	Opus	Geometric primitives (CPU and GPU); exact fast backend; proteome run; ablations; speed work	Exactness tests against the reference; dev reviewer (20)
6–11	Design reviewers, round 1	Opus ×6	reviews/round1/*.md (external PI; prior work and data; statistics; algorithms and compute; AI-agent failure modes; biophysics and AlphaFold)	Changes mapped in DESIGN.md §8
12–13	Design reviewers, round 2	Opus ×2	reviews/round2/*.md (external PI and statistics; data and agent risks)	Changes in DESIGN.md §9
14	Training and dev set builder	Opus	Leakage-controlled BioLiP2 training and dev sets; apo counterparts (docs/TRAINSET.md)	Leakage auditor (19): marginal fail, 186 entries removed
15	AlphaFold test set and AlphaFold training labels	Opus	AlphaFold benchmark (docs/AF_BENCHMARK.md); AlphaFold training and dev labels	Test verifier (21) for O5
16	Dimer benchmark builder	Opus	Dimer benchmark and evaluator (docs/DIMER_BENCHMARK.md)	Verification script; external report reviewer (25) found a stratum bug
17	Method developer	Opus	BURROW reference implementation, training, presets, v2 and v3 cycles, packaging (docs/METHOD.md)	Dev reviewer (20); test verifier (21)
18	AlphaFold-mode developer	Opus	AlphaFold-mode mechanisms, ablations, optional AlphaFold ranker (docs/AF_MODE.md)	Pre-registered pLDDT-filter control; dev reviewer (20)
19	Independent leakage auditor	Sonnet	results/audit/LEAKAGE_AUDIT.md	–
20	Independent dev-result reviewer	Opus	results/review_dev/REVIEW.md	–
21	Independent test verifier	Opus	results/verify_test/VERIFY.md (all headline values within 0.2 points)	–
22–23	Process retrospectives	Sonnet ×2	reviews/process/M2_retro.md, FINAL_retro.md	–
24	Report writer and number checker	Opus	REPORT.md; scripts/check_report_numbers.py	External report reviewer (25)
25	External report reviewer	Opus	reviews/report/REPORT_REVIEW.md	–
26	Manuscript writer	Opus	This manuscript (manuscript/)	Submitting authors
27	Figure maker	[to be confirmed]	manuscript/figures/ (every plotted value read from committed result files)	Submitting authors